

Problem 15.4.3

Find the Maclaurin series and its radius of convergence.

$$\sin 2z^2$$

Solution

$$\sin(w) = \sum_{n=1}^{\infty} (-1)^n \frac{w^{2n+1}}{(2n+1)!}$$

$$\sin 2z^2 = \sum_{n=1}^{\infty} (-1)^n \frac{(2z^2)^{2n+1}}{(2n+1)!} = \boxed{\sum_{n=1}^{\infty} (-1)^n \frac{2^{2n+1} z^{4n+2}}{(2n+1)!}}$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{(-1)^n} \cdot \frac{2^{2n+3}}{2^{2n+1}} \cdot \frac{(2n+1)!}{(2n+3)!} \right| \\ &= \lim_{n \rightarrow \infty} \left| -1 \cdot 4 \cdot \frac{1}{(2n+3)(2n+2)} \right| \\ &= 0 \end{aligned}$$

$$\therefore R = \boxed{\infty}$$

Problem 15.4.16. Inverse sine.

Developing $1/\sqrt{1-z^2}$ and integrating, show that

$$\sin^{-1} z = z + \left(\frac{1}{2}\right) \frac{z^3}{3} + \left(\frac{1 \cdot 3}{2 \cdot 4}\right) \frac{z^5}{5} + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right) \frac{z^7}{7} + \cdots (|z| < 1)$$

Solution

$$f(z) = \sin^{-1} z$$

$$f'(z) = (1 - z^2)^{-1/2}$$

$$(1 + w)^{-m} = \sum_{n=0}^{\infty} \binom{-m}{n} w^n = 1 - mw + \frac{m(m+1)}{2!} w^2 - \frac{m(m+1)(m+2)}{3!} w^3 + \cdots$$

$$\begin{aligned} f'(z) &= \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} (-z^2)^n \\ &= 1 - \frac{1}{2} (-z^2) + \frac{\frac{1}{2}(\frac{1}{2}+1)}{2!} (-z^2)^2 - \frac{\frac{1}{2}(\frac{1}{2}+1)(\frac{1}{2}+2)}{3!} (-z^2)^3 + \cdots \\ &= 1 + \frac{1}{2} z^2 + \frac{\frac{1}{2}(\frac{3}{2})}{2} z^4 + \frac{\frac{1}{2}(\frac{3}{2})(\frac{5}{2})}{6} z^6 + \cdots \\ &= 1 + \left(\frac{1}{2}\right) z^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right) z^4 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right) z^6 + \cdots \end{aligned}$$

Integrating termwise

$$f(z) = z + \left(\frac{1}{2}\right) \frac{z^3}{3} + \left(\frac{1 \cdot 3}{2 \cdot 4}\right) \frac{z^5}{5} + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right) \frac{z^7}{7} + \cdots$$

This converges on $|z| < 1$ due to the geometric series f'

Problem 15.4.19

Find the Taylor series with center $z_0 = i$ and its radius of convergence.

$$\frac{1}{1-z}$$

Solution

$$\begin{aligned} f(z) &= \frac{1}{1-z} \\ &= \frac{1}{1-i-(z-i)} \\ &= \frac{1}{(1-i)\left(1-\frac{z-i}{1-i}\right)} \\ &= \frac{1}{1-i} \sum_{n=0}^{\infty} \left(\frac{z-i}{1-i}\right)^n \quad \text{by 15.4(11)} \end{aligned}$$

This series converges on

$$\begin{aligned} \left| \frac{z-i}{1-i} \right| &< 1 \\ |z-i| &< |1-i| \\ |z-i| &< \sqrt{2} \\ \therefore R &= \boxed{\sqrt{2}} \end{aligned}$$

Problem 15.4.21

Find the Taylor series with center $z_0 = \frac{\pi}{2}$ and its radius of convergence.

$$\sin z$$

Solution

$$\begin{aligned} f(z) &= \sin z & f(z_0) &= 1 \\ f'(z) &= \cos z & f'(z_0) &= 0 \\ f''(z) &= -\sin z & f''(z_0) &= -1 \\ f^{(3)}(z) &= -\cos z & f^{(3)}(z_0) &= 0 \\ f^{(4)}(z) &= \sin z & f^{(4)}(z_0) &= 1 \end{aligned}$$

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} a_n (z - z_0)^n & a_n &= \frac{1}{n!} f^{(n)}(z_0) \\ &= 1 + 0 - \frac{1}{2!} \left(z - \frac{\pi}{2}\right)^2 + 0 + \frac{1}{4!} \left(z - \frac{\pi}{2}\right)^4 + \cdots \\ &= 1 - \frac{1}{2!} \left(z - \frac{\pi}{2}\right)^2 + \frac{1}{4!} \left(z - \frac{\pi}{2}\right)^4 - \cdots \\ &= \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(z - \frac{\pi}{2}\right)^{2n}} \end{aligned}$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ L &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{(-1)^n} \right| \\ L &= \lim_{n \rightarrow \infty} \left| \frac{-1}{(2n+2)(2n+1)} \right| \\ L &= 0 \\ \therefore R &= \boxed{\infty} \end{aligned}$$

Problem 15.4.23

Find the Taylor series with center $z_0 = i$ and its radius of convergence.

$$\frac{1}{(z+i)^2}$$

Solution

$$\begin{aligned} f(z) &= \frac{1}{(z+i)^2} & f(z_0) &= -\frac{1}{4} & a_0 &= -\frac{1}{4} \\ f'(z) &= -\frac{2}{(z+i)^3} & f'(z_0) &= -\frac{i}{4} & a_1 &= -\frac{i}{4} \\ f''(z) &= \frac{6}{(z+i)^4} & f''(z_0) &= \frac{3}{8} & a_2 &= \frac{3}{16} \\ f^{(3)}(z) &= -\frac{24}{(z+i)^5} & f^{(3)}(z_0) &= \frac{3i}{4} & a_3 &= \frac{i}{8} \\ & & & & a_n &= -\frac{n+1}{4} \left(\frac{i}{2}\right)^n \end{aligned}$$

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} a_n (z - z_0)^n \\ &= \boxed{-\sum_{n=0}^{\infty} \frac{n+1}{4} \left(\frac{i}{2}\right)^n (z-i)^n} \end{aligned}$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ L &= \lim_{n \rightarrow \infty} \left| \frac{n+2}{4} \left(\frac{i}{2}\right)^{n+1} \cdot \frac{4}{n+1} \left(\frac{i}{2}\right)^{-n} \right| \\ L &= \lim_{n \rightarrow \infty} \left| \frac{n+2}{n+1} \left(\frac{i}{2}\right) \right| \\ L &= \left| \frac{i}{2} \right| \\ L &= \frac{1}{2} \\ \therefore R &= \boxed{2} \end{aligned}$$